Solving systems of equations with the elimination method

1

a 7 **b** $x = -\frac{125}{17}$ **c** $y = \frac{39}{17}$

2

a $2x + 3y = -7$ and $-9x + y = -4$ **b** $x = \frac{5}{29}$

3

a $-29x = 6$ **b** $x = -\frac{6}{29}, y = \frac{48}{29}$

4 When we combine the equations, all the variables are eliminated and we end up with a false statement such as $0 = 2$.

5

a

i Infinitely many solutions **ii** Dependent

b

i No solutions **ii** Inconsistent

6 Multiply by 5

7

a Multiply the first equation by 7 and multiply the second equation by 6, and then subtract the equations.

b Multiply the first equation by 10 and multiply the second equation by 100 to remove the decimals. Then add the equations.

8

a $x = 5, y = 5$ **b** $x = -9, y = 4$ **c** $x = 6, y = 7$ **d** $x = 2, y = \frac{1}{5}$

e $x = -20, y = 15$ **f** $x = \frac{1}{2}, y = \frac{3}{2}$ **g** $p = \frac{3}{5}, q = \frac{4}{5}$

9

a -1 **b** 59 **c** -11

© ClipSAT: By Mr. Mohamed Abdallah +966597688647